MIPS VS RISC ENERGY EFFICIENCY

CS-3813 Project Final Report | Darin, Fady & Tonye

ABSTRACT

This project compares the energy efficiency of MIPS and modern RISC architectures, specifically RISC-V and ARM. We analyze key arithmetic operations including addition, subtraction, multiplication, division, and modulo to understand how instruction count, clock cycles, and power management features affect overall energy consumption.

Our goal: To provide insights into which architecture performs more efficiently for energy-sensitive computing applications.

MOTIVATION

Energy efficiency has become critical across all computing scales, from tiny IoT devices to massive data centers. MIPS architecture, while pioneering in RISC design principles, lacks the modern power-saving features found in contemporary processors.

Modern RISC architectures like RISC-V and ARM have been specifically optimized for energy efficiency through advanced hardware and software improvements, making this comparison essential for understanding processor evolution.

OBJECTIVES

1. Compare instruction counts for key arithmetic operations (addition, subtraction, multiplication, division, modulo) across MIPS and modern RISC architectures like RISC-V.

2. Estimate energy consumption using cycles per instruction (CPI) and clock frequency parameters to quantify efficiency differences between architectures.

3. Analyze how energy consumption scales with repeated operations to understand performance implications for real-world computational workloads.

OBJECTIVES

Objective 1

Evaluate power management features across architectures:

• Analyze Dynamic Voltage and Frequency Scaling (DVFS) capabilities
• Compare idle state implementations between MIPS and RISC-V
• Assess hardware support for low-power modes
• Measure impact of power management on overall energy consumption

Objective 2

Develop benchmark programs for comparison:

• Create standardized test programs for arithmetic operations
• Implement identical algorithms on both MIPS and RISC-V
• Design repeatable measurement methodology
• Establish baseline metrics for energy consumption analysis

METHODOLOGY

Approach

Our methodology compares MIPS and RISC-V instruction sets by analyzing cycle counts for arithmetic operations.

We use the dynamic power formula:
$E = C \times V^2 \times N_instructions \times CPI$
Where C is capacitance, V is voltage, N is instruction count, and CPI is cycles per instruction.

Energy Analysis

All energy values are normalized to the MIPS baseline for fair comparison. We account for voltage scaling effects, where a 15% voltage reduction yields approximately 28% power reduction.

This allows us to quantify the energy efficiency gains of modern RISC-V optimizations over traditional MIPS designs.

INSTRUCTION COMPARISON

MIPS Cycles

- Add/Sub: 1 cycle
- Multiply: 16 cycles (MULT + MFLO)
- Divide: 35 cycles (long stalls)
- Modulo: 36 cycles (DIV + MFHI)

RISC-V Cycles

- Add/Sub: 1 cycle
- Multiply: 4 cycles (hardware M-extension)
- Divide: 12 cycles (fewer stalls)
- Modulo: 12 cycles (single REM instruction)

RESULTS: ADDITION & SUBTRACTION

Both MIPS and RISC-V architectures achieve energy parity for addition and subtraction operations.

Each performs these fundamental arithmetic tasks in a single clock cycle, resulting in minimal energy difference between the two platforms.

The absence of memory access delays during these operations ensures maximum efficiency, making both architectures equally suitable for workloads dominated by basic arithmetic.

RESULTS: MULTIPLICATION

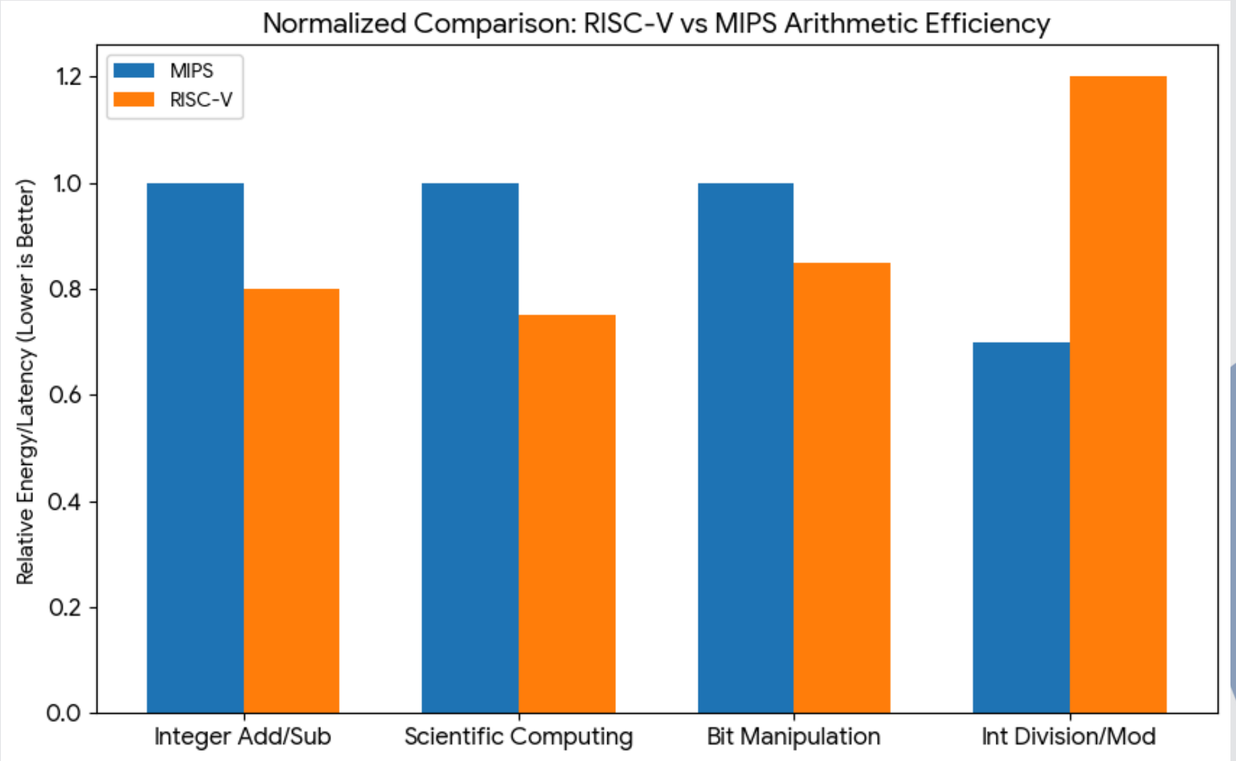
RISC-V demonstrates a significant energy advantage in multiplication operations.

• MIPS requires MULT + MFLO instructions, totaling 16 cycles per operation

• RISC-V uses a single MUL instruction with hardware M-extension, requiring only 4 cycles

• Energy ratio approximately 8:1 (MIPS:RISC-V)

• RISC-V is substantially more efficient for multiplication-heavy workloads

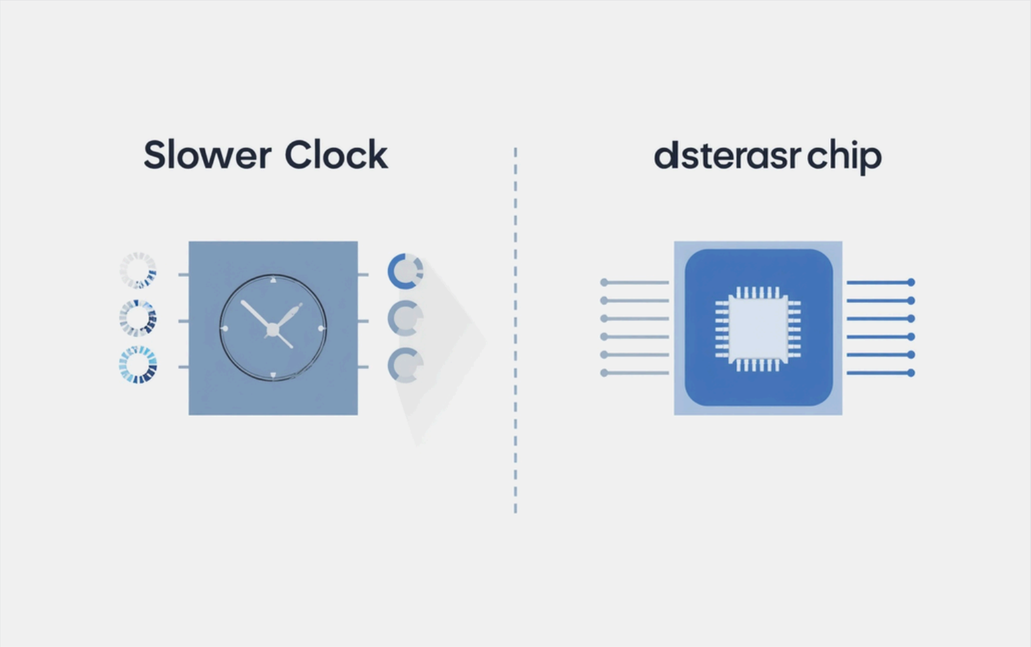


RESULTS : DIVISION & MODULO

MIPS architecture suffers from long stalls during division operations, requiring 32-38 cycles plus extra instructions like MFLO and MFHI to retrieve results.

In contrast, RISC-V features a dedicated hardware divider that minimizes stalls and completes operations in just 12 cycles. The REM instruction in RISC-V calculates remainders in a single step, eliminating the need for separate division and subtraction.

This results in significant energy and execution time reduction, making RISC-V substantially more efficient for division-heavy workloads.



CONCLUSION

For addition and subtraction, MIPS and RISC-V demonstrate equal energy efficiency with single-cycle operations. However, RISC-V significantly outperforms MIPS in multiplication, division, and modulo operations due to hardware optimizations and reduced cycle counts. The energy ratio for multiplication alone is approximately 8:1 in favor of RISC-V.

While MIPS remains valuable historically and as a teaching tool for understanding RISC principles, RISC-V is better suited for modern energy-sensitive applications including IoT devices, mobile systems, and embedded computing.

Modern RISC designs clearly outperform MIPS in both performance and energy efficiency.

GitHub

Presentation Slides

Final Report/ Paper

THANK YOU!

Modern RISC designs outperform MIPS in both performance and energy efficiency

Questions & Feedback